

Week4 Task - SQL knowledge

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- **First, second, and third normal forms:** Rules to organize database tables to reduce data redundancy and dependency (1NF: atomic values, 2NF: no partial dependency, 3NF: no transitive dependency).
- **Primary, foreign, composite primary key:** Unique identifiers for rows (Primary), links between tables (Foreign), and keys made of two or more columns combined (Composite).
- **Normalization:** The process of structuring a relational database to minimize data duplication.
- **Relational database:** A database that stores data in tables with rows and columns, linked by relationships.
- **SSMS:** SQL Server Management Studio, the official GUI tool to manage SQL Server instances.
- **MDF, LDF, NDF, BAK files:** Database file types representing Primary data (MDF), Transaction log (LDF), Secondary data (NDF), and Backup (BAK).
- **Windows and SQL authentication:** Two methods to log into SQL Server: using Windows credentials or SQL Server-specific usernames/passwords.
- **ER diagrams, entities, attributes, and relationships:** Visual modeling tools showing database structures (Entities = tables, Attributes = columns, Relationships = connections).
- **Linked server vs PolyBase:** Methods to query external data sources from SQL Server (Linked Server for traditional RDBMS; PolyBase for big data like Hadoop/Azure).
- **DB security (best practices):** Guidelines to protect databases (e.g., least privilege access, encryption, strong passwords).
- **Queries – create, select, insert, update, delete, etc.:** Commands used to define, retrieve, and manipulate data in SQL.
- **SQL joins – inner, left, right, full:** Operations to combine rows from multiple tables based on related columns.
- **Index:** A database structure that improves the speed of data retrieval operations.
- **Performance tuning – table scan, cluster index, scan, query plan, etc.:** Techniques and tools to optimize query execution speed and resource usage.
- **Stored procedures:** Precompiled collections of SQL statements stored on the server for reuse.
- **SQL server job scheduler:** A tool (SQL Server Agent) used to automate repetitive administrative tasks on a schedule.
- **SQL functions:** Routines that accept parameters, perform calculations or actions, and return a result.
- **Trigger:** A special stored procedure that automatically executes in response to specific database events (e.g., INSERT, UPDATE).
- **Transactions:** A sequence of SQL operations executed as a single unit (ACID properties: Atomicity, Consistency, Isolation, Durability).
- **Disaster recovery:** Strategies and procedures to restore database operations after a catastrophic failure.
- **Replication:** The process of copying and distributing data from one database to another for synchronization.
- **AAG (always on available group):** High-availability and disaster-recovery solution providing enterprise-level failover environments.
- **Temporal table:** A system-versioned table that automatically tracks the full history of data changes over time.